derivatives $\frac{d}{dx}(x^n) = nx^{n-1}$

Basics / 1st Principle

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{\Delta (x^n)}{\Delta x} \text{ ------}$$

⊛ Fuck to approach any Schema Design in general.

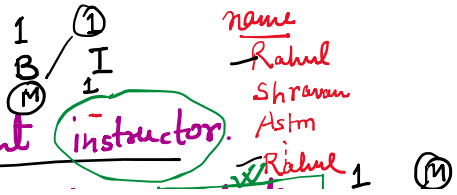
⊛ Schema (wrt DB).

- St. of DB
- Tables in DB → main tables, Index", Rel^{n}ship tables.
- Col. in DB
- Key → Pk, Fk
- Index
- Pictorial Represent^n

⊛ How to approach a Schema Design?

Name of DB Design AirTribe

1) AirTribe will have multiple batches.
2) For each batch, we need to store the name, start month and current instructor.



2.) For each batch, we need to store the name, start month and current instructor.

3.) Each batch of AirTribe will have multiple students.

4.) Each batch has multiple classes.

5.) For each class; store the name, date & time, instructor of the class.

6.) For every student, we store their name, grad yr., university name, email, phoneNo,

7.) Every student has a buddy, who is also a student.

8.) A student may move from one batch to another. (history to be maintained)

9.) For each batch a student moves to, the date of starting is stored.

10.) Every student has a mentor.

11.) For every mentor, we store their name & current company.

12.) Store info. about mentorship sessions (time, duration, student, mentor, student rating, mentor rating)

13.) For every batch, we store if its CoreSWE-batch or AI/ML-batch.

can or can not, take a decision ↓ justify it.

✳ Steps to Follow :-

1. Create the Tables :-

- Find out all the nouns in the req.
- After identifying the noun, do we need to store any data related to the noun ?

↓

If Yes → create a Table

Naming conventions
- plural names
- snake-case
- Attributes (Cols.) in singular forms

Avoid redundancy & have more consistency by creating a new table
— Ex. - Instructors.

② Add PK & all other attributes

Expectations from a PK
↳ unique, not-null
↳ small in size [int >>>>> string] preference-wise
↳ data type matters
↳ It should rarely change.
↳ Index also changes (⇒ entire index B+Tree changes)
↳ Don't choose a PK which is not sys. generated

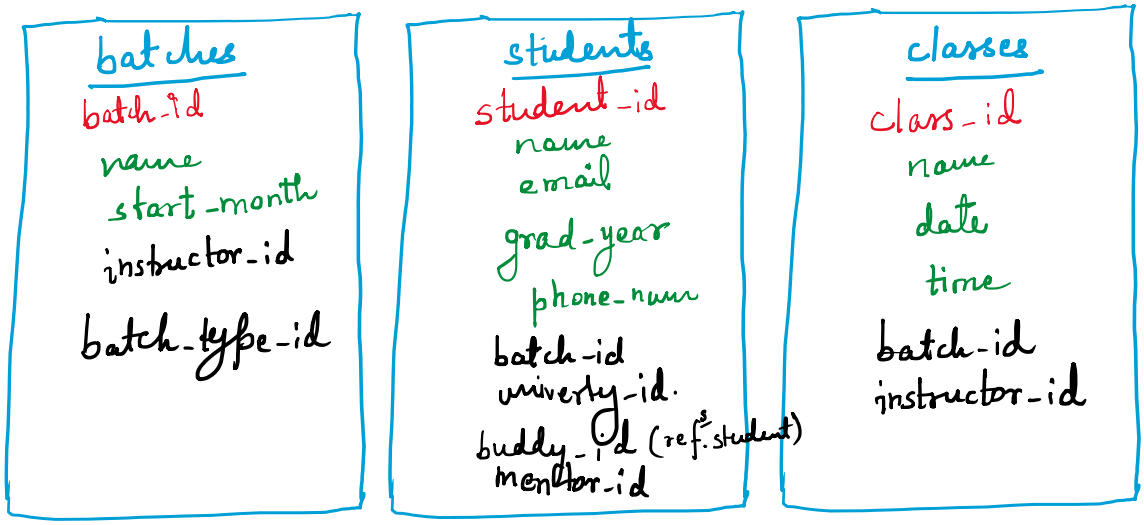
∴ use sys-generated IDs as PK.

Don't think about rel.ship attributes.

③ → Identify the relationships (By the 2-step process)
↳ Add as attribute → 1:1
↳, 1:M / M:1

↳ Add as attribute → 1:1
↳ 1:M / M:1

or create a new rel'ship table → M:M


batches
batch_id
name
start_month
instructor_id
batch_type_id
students
student_id
name
email
grad_year
phone_num
batch_id
univerty_id.
buddy_id (ref. student)
mentor_id
classes
class_id
name
date
time
batch_id
instructor_id


instructors
instructor_id
name
mentors
mentor_id
name
organization_id
mentor_sessions
mentor_session_id
time
duration
student_rating
mentor_rating
student_id
mentor_id
organizations
organization_id
organizn_name
universities
university_id
university_name
batch_history
recid (PK)
student_id
batch_id
starting_date
batch_type
batch_type_id
batch_name

⑧ How to store batch types

↳ String / varchar → typo-errors.
→ huge space
→ Update leading to data corruption / inconsistency

↓

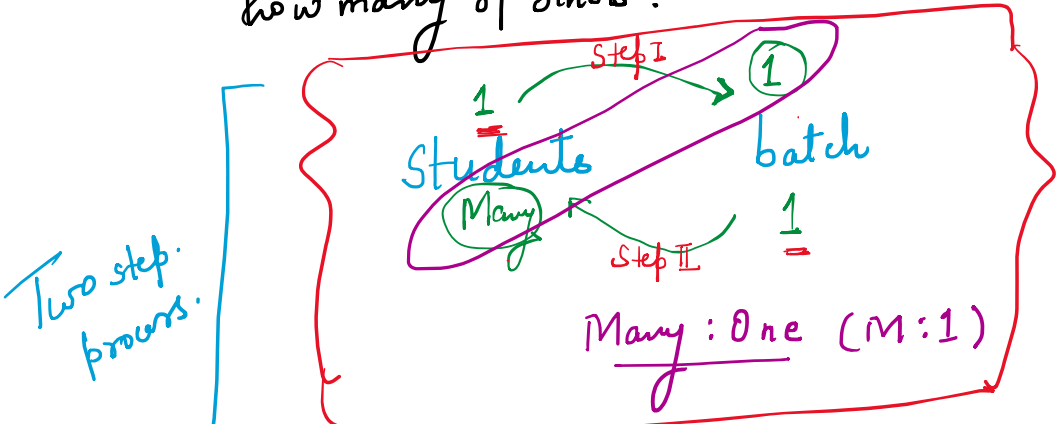
Sol^n

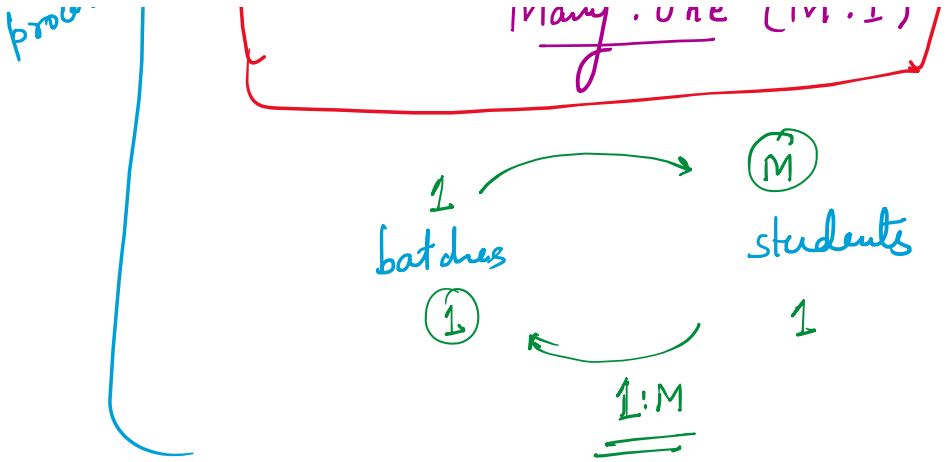
↓

Store id's of diff. batch-types

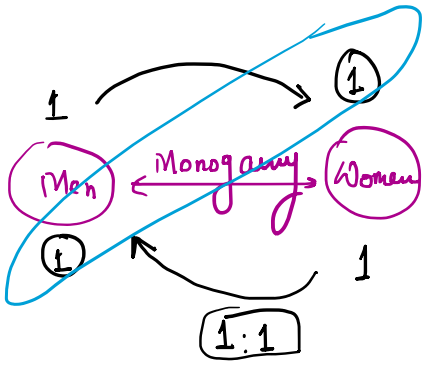
Cardinality

When two things are related to each other, how many of one thing is related to how many of others.

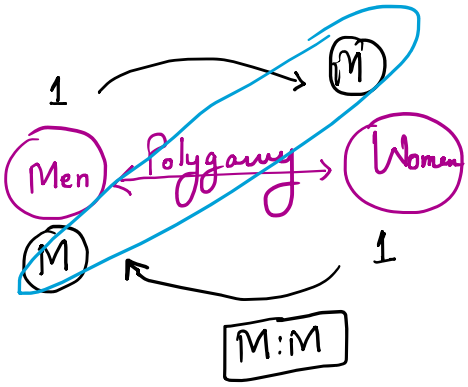




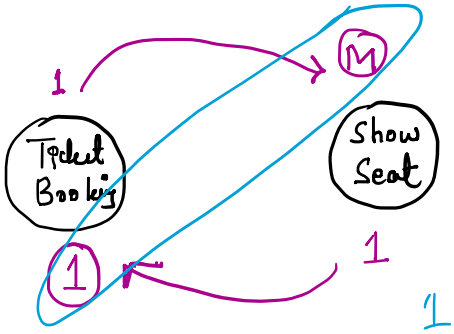
1.) 1 : 1



2.) M : M



3.) M : 1 or 1 : M



Show
- Screen
- Time
- Movie
- Theatre

Similarly

↳ Similarly

ShowSeat v/s TicketBooking
⟷ M:1

Q.1 Facebook → Friends

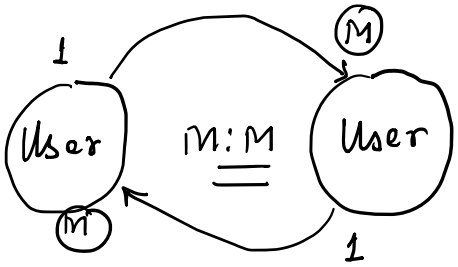
Q.2 Order & Product (B2C)

Q.3 Student & Course

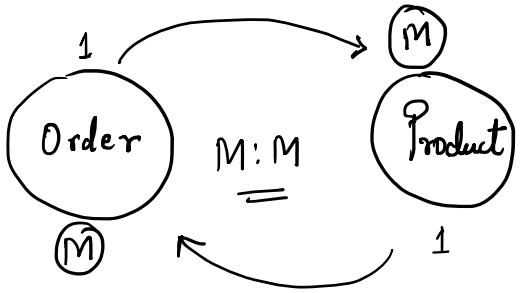
Q.4. Customer & Account

Q.5 Student & Major

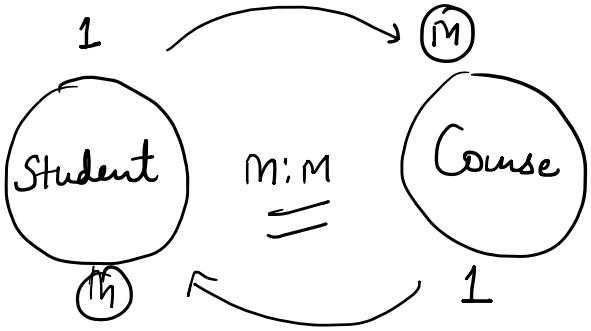
A.1



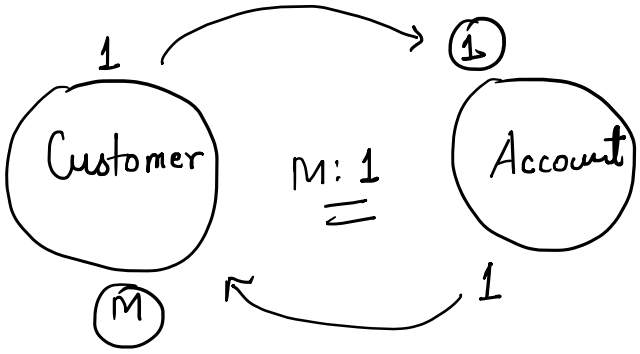
A.2



A.3.

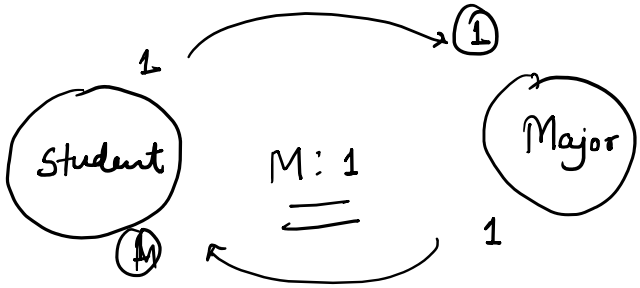


A.4.



Joint

A.5



⊛ How to represent cardinalities in tables ?

1.) 1:1 → Choose any one of the table, where you will keep the id of the other.

Men

id	name	Woman-id
1	A	2
2	B	3
3	C	1

Women

id	name	Men-id
1	X	3
2	Y	1
3	Z	2

Is it possible

men

id	name	----	Woman-2	name	-----
1	A	---	2	Y	- -
2	B	---	3	Z	- —
3	C	---	1	X	----

men table then has woman info/ data

Is it possible

monogamy

m-id	w-id
1	2
2	3
3	1

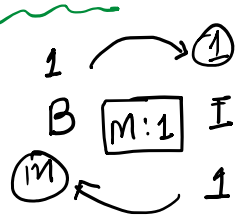
Fine, but unnecessarily extra table created

unnecessary extra (table) created

How do you find complete info of a woman related to a man?

JOIN

2.) 1:M or M:1 → put id of '1' side to the 'M' side.



B (M:1) I

Batches

id	name	I-id
2		401
3		401

Instructors

id	name	b-id
401		[2, 3, ...]

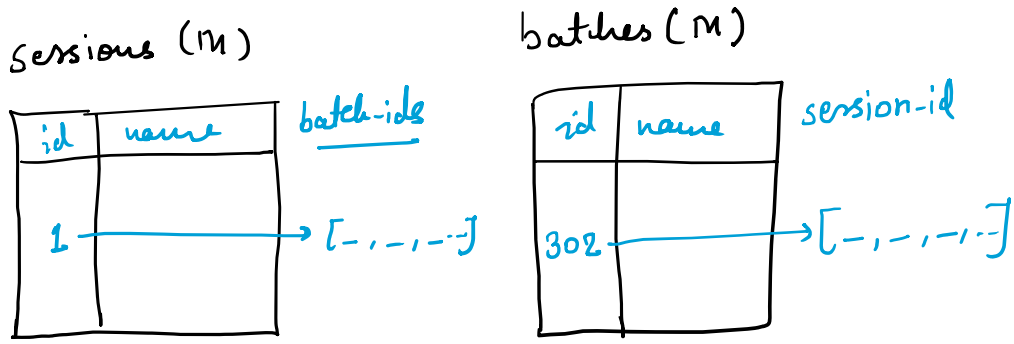
batches-instructors

b-id	i-id
2	401
3	401

Note could have created a sep. rel^ship table → unnecessarily wasting space

3.) M:M

Sessions (M:M) Batches



↓ create a mapping table

batches_sessions

batch_id	session_id
1	301
1	302
1	303
2	2
2	302

Resume

Internal working of SQL Queries

10:17 pm

10:16 pm

(*) departments

dept_id (PK), dept_name

(*) employees

emp_id (PK), firstName, lastName,
hire_date, salary, dept_id (FK)

(*) projects

project_id (PK), projectName, manager_id.

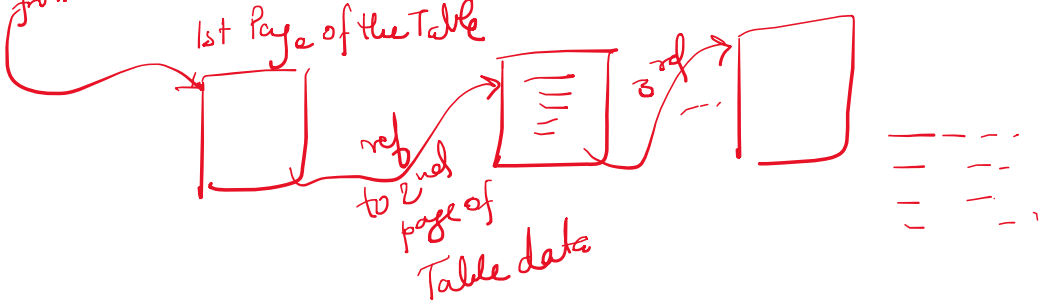
Select * from employees;

Full TableScan

~~Full Disk Scan~~

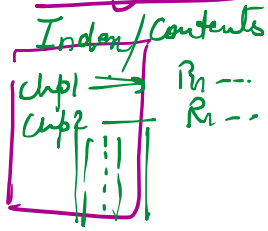
exec sp_help 'emp';

ref. extracted from metadata in the DB



(*) Always created on Attributes | grp. of attributes

⊛ Indexing / Indices





multiple pages

You want to read Ch#4 for Part 5

Alt-1 → Go to Index → Take the page no. & directly go over there

≈ O(1)

Alt-2 → Flip page one after other to find the Ch#4

≈ O(N)

Alt 1 is much faster than Alt-2

Similarly

In DB's also

Index → Clustered Index (default)

→ Non-clustered Index (explicit, needs to "

(explicit
needs to
be created)

Clustered Index

default Index on PK.

B+ Tree

SBBST aka AVL Tree